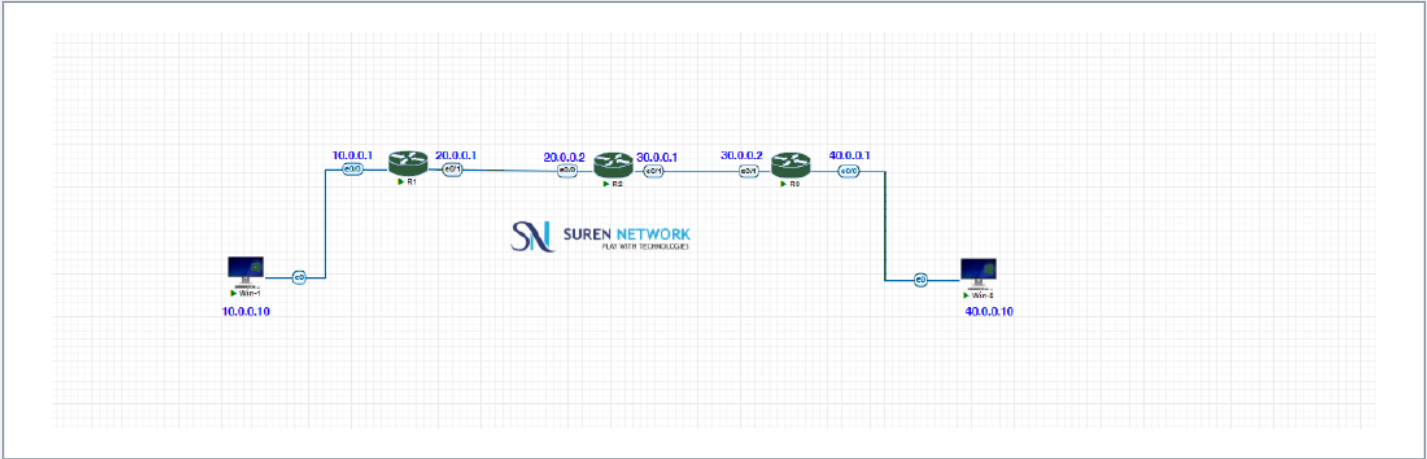
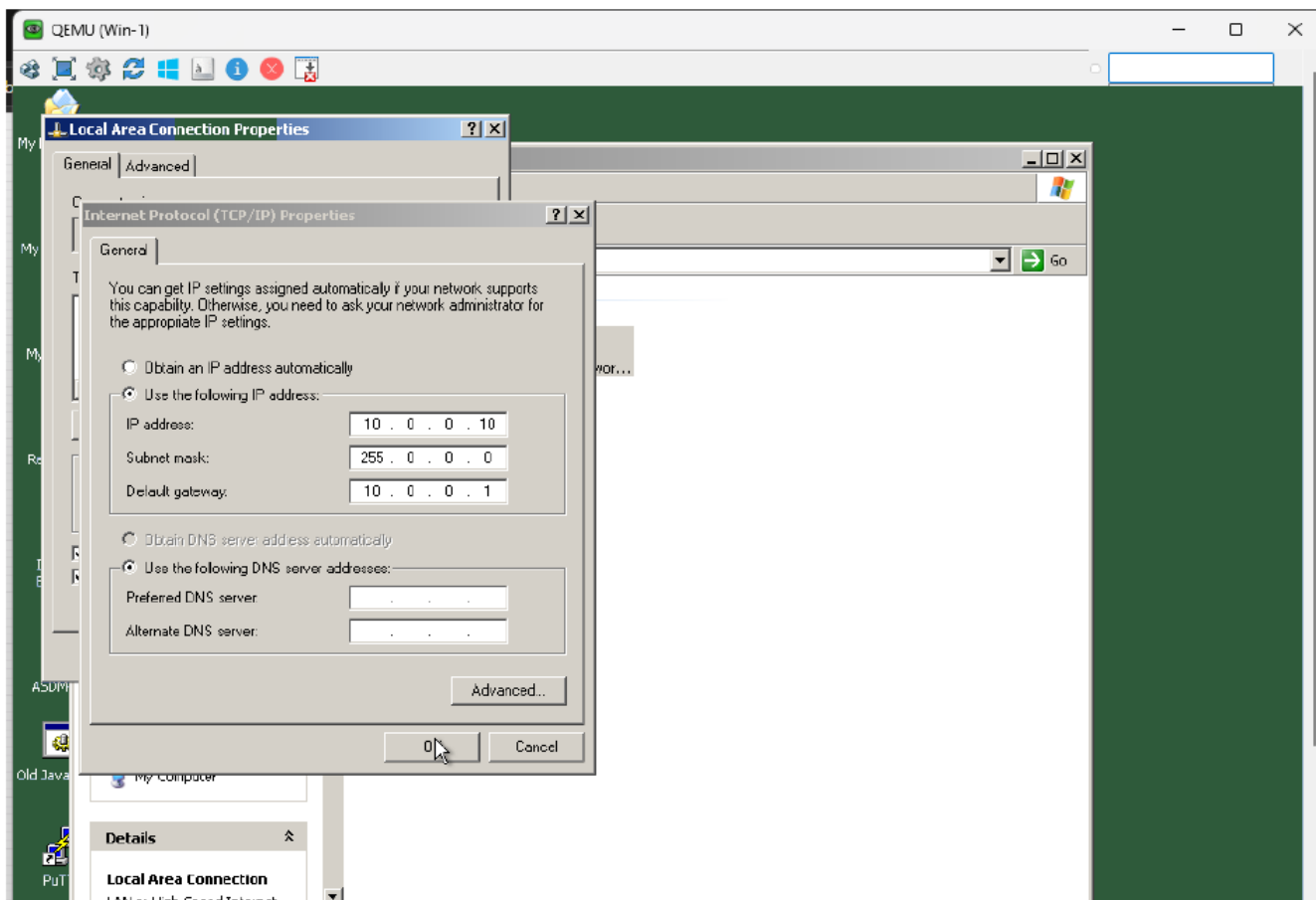




This is the lab topology: Win-1 and Win-2 are end devices connected through routers R1, R2, and R3.



Win-1 is given a static IP (10.0.0.10) with gateway 10.0.0.1 so it can reach R1.

```
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
Ethernet0/0              unassigned      YES unset  administratively down  down
Ethernet0/1              unassigned      YES unset  administratively down  down
Ethernet0/2              unassigned      YES unset  administratively down  down
Ethernet0/3              unassigned      YES unset  administratively down  down
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface ethernet 0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut down
                        ^
% Invalid input detected at '^' marker.

Router(config-if)#no shutdown
Router(config-if)#exit
*Aug 15 14:48:10.396: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Aug 15 14:48:11.396: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config-if)#exit
```

R1's Ethernet0/0 is assigned 10.0.0.1 and enabled with 'no shutdown'.

```
Router(config-if)#no shutdown
Router(config-if)#exit
Aug 15 14:48:10.396: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
Aug 15 14:48:11.396: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#
Aug 15 14:48:55.865: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
Ethernet0/0              10.0.0.1        YES manual up                    up
Ethernet0/1              unassigned      YES unset administratively down down
Ethernet0/2              unassigned      YES unset administratively down down
Ethernet0/3              unassigned      YES unset administratively down down
Router#
```

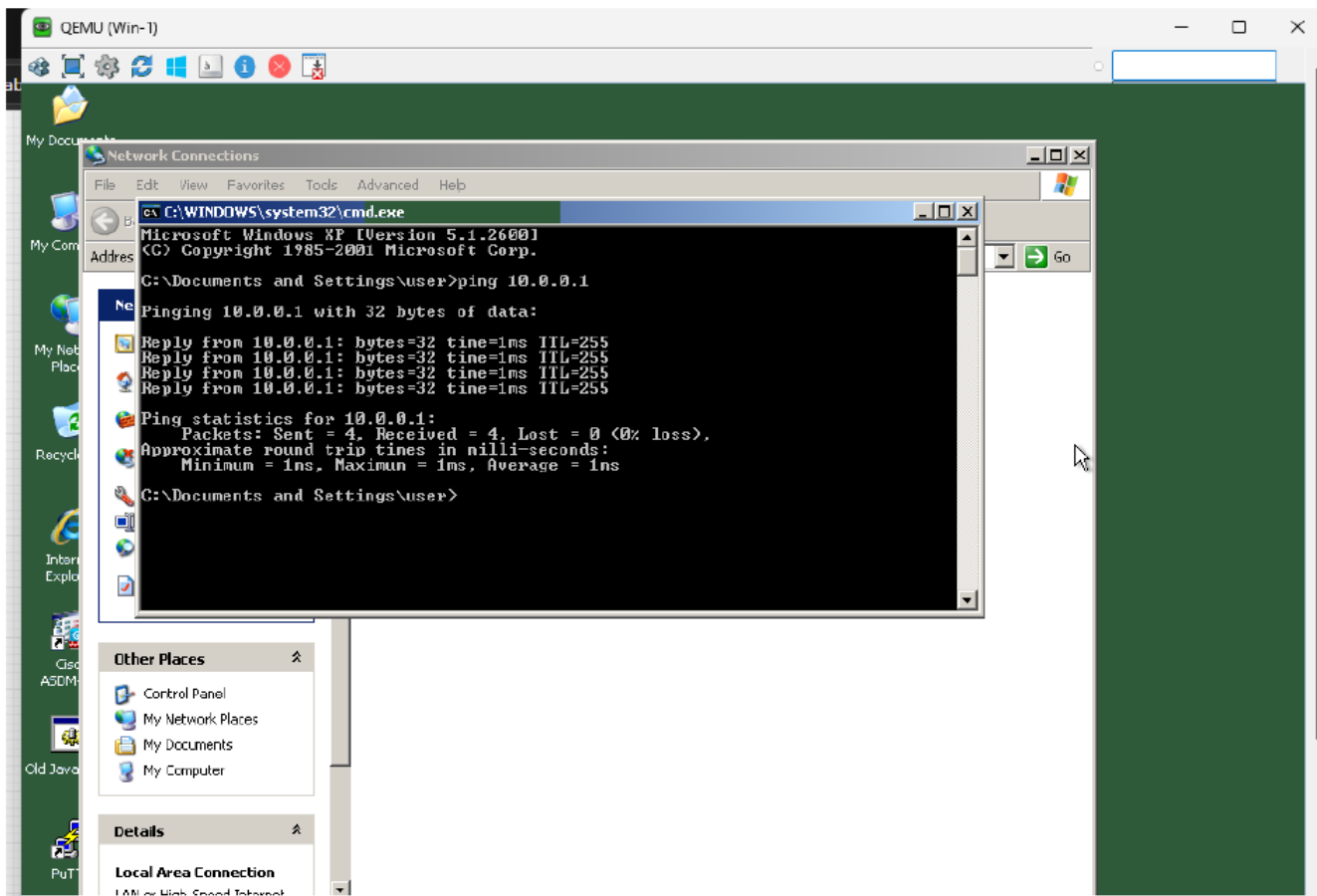
'show ip interface brief' confirms Ethernet0/0 is up with IP 10.0.0.1.

```
Router#
*Aug 15 14:48:55.865: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              10.0.0.1        YES manual up          up
Ethernet0/1              unassigned      YES unset   administratively down down
Ethernet0/2              unassigned      YES unset   administratively down down
Ethernet0/3              unassigned      YES unset   administratively down down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface ethernet 0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
*Aug 15 14:52:15.625: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Aug 15 14:52:16.625: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
```

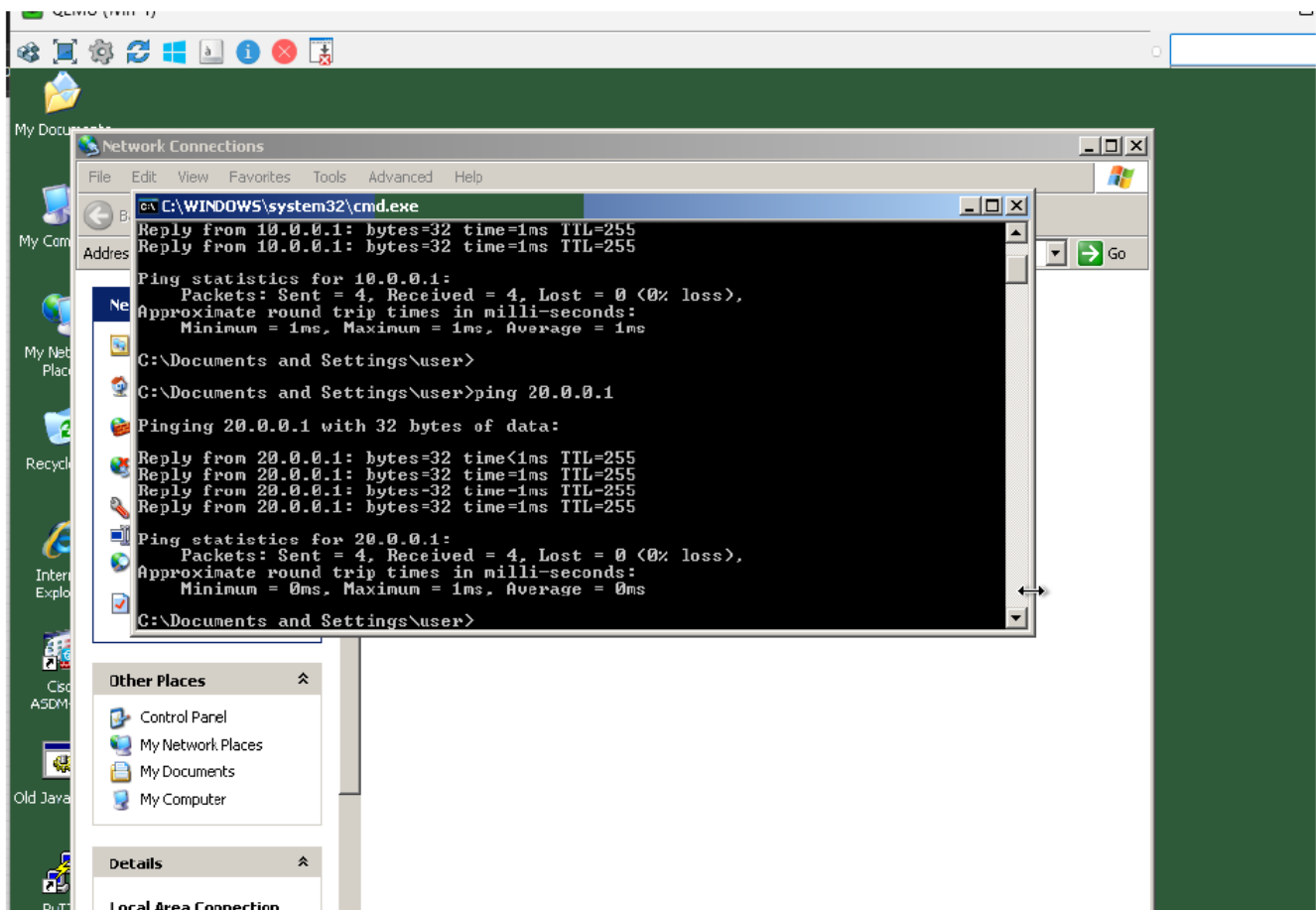
R1's Ethernet0/1 is assigned 20.0.0.1 and enabled, linking R1 to R2.

```
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface ethernet 0/1
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
*Aug 15 14:52:15.625: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Aug 15 14:52:16.625: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config-if)#exit
Router(config)#exit
Router#s
*Aug 15 14:52:53.865: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    10.0.0.1        YES manual up          up
Ethernet0/1    20.0.0.1        YES manual up          up
Ethernet0/2    unassigned      YES unset  administratively down down
Ethernet0/3    unassigned      YES unset  administratively down down
Router#
```

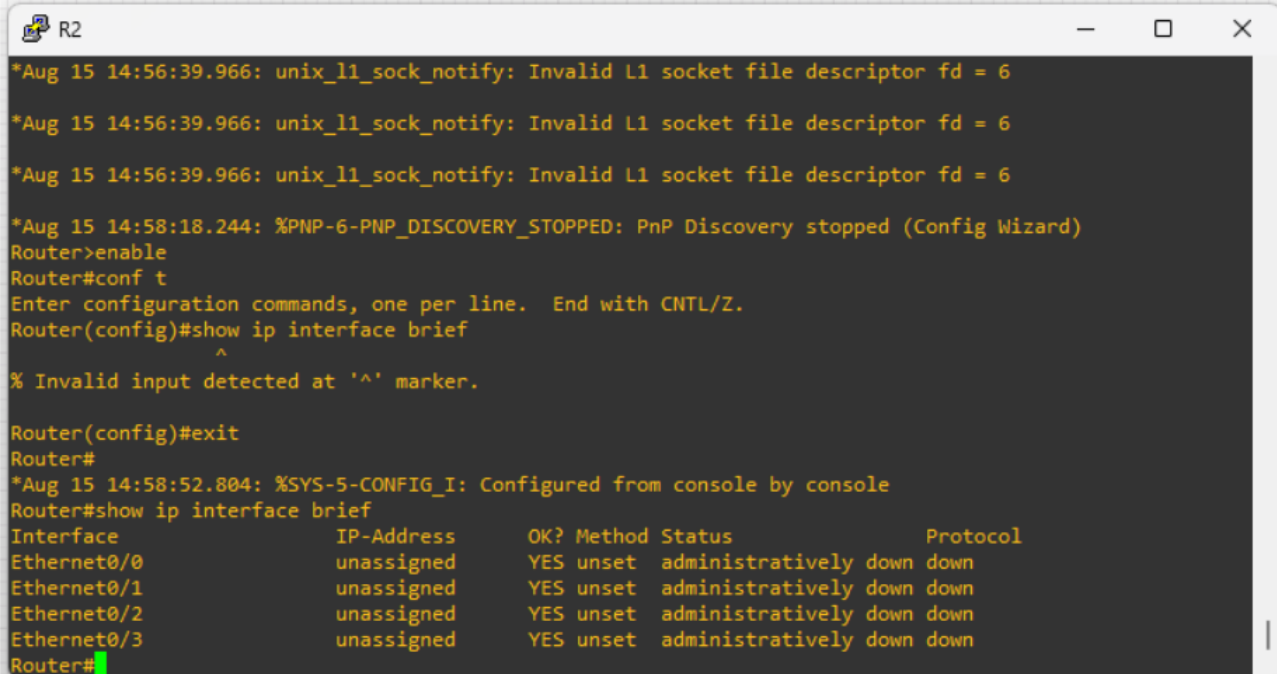
Both R1 interfaces are now up: 10.0.0.1 and 20.0.0.1.



Win-1 pings its gateway 10.0.0.1 successfully, confirming local connectivity.



Win-1 also pings 20.0.0.1 successfully, reaching across R1.



```
*Aug 15 14:56:39.966: unix_l1_sock_notify: Invalid L1 socket file descriptor fd = 6
*Aug 15 14:56:39.966: unix_l1_sock_notify: Invalid L1 socket file descriptor fd = 6
*Aug 15 14:56:39.966: unix_l1_sock_notify: Invalid L1 socket file descriptor fd = 6
*Aug 15 14:58:18.244: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show ip interface brief
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
*Aug 15 14:58:52.804: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
Ethernet0/0              unassigned      YES unset  administratively down  down
Ethernet0/1              unassigned      YES unset  administratively down  down
Ethernet0/2              unassigned      YES unset  administratively down  down
Ethernet0/3              unassigned      YES unset  administratively down  down
Router#
```

R2 starts with no interfaces configured yet.

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#show ip interface brief
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
*Aug 15 14:58:52.804: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
Ethernet0/0              unassigned      YES unset  administratively down  down
Ethernet0/1              unassigned      YES unset  administratively down  down
Ethernet0/2              unassigned      YES unset  administratively down  down
Ethernet0/3              unassigned      YES unset  administratively down  down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface ethernet0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Aug 15 15:00:56.224: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Aug 15 15:00:57.224: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config)#interface ethernet0/1
      ^
% Invalid input detected at '^' marker.

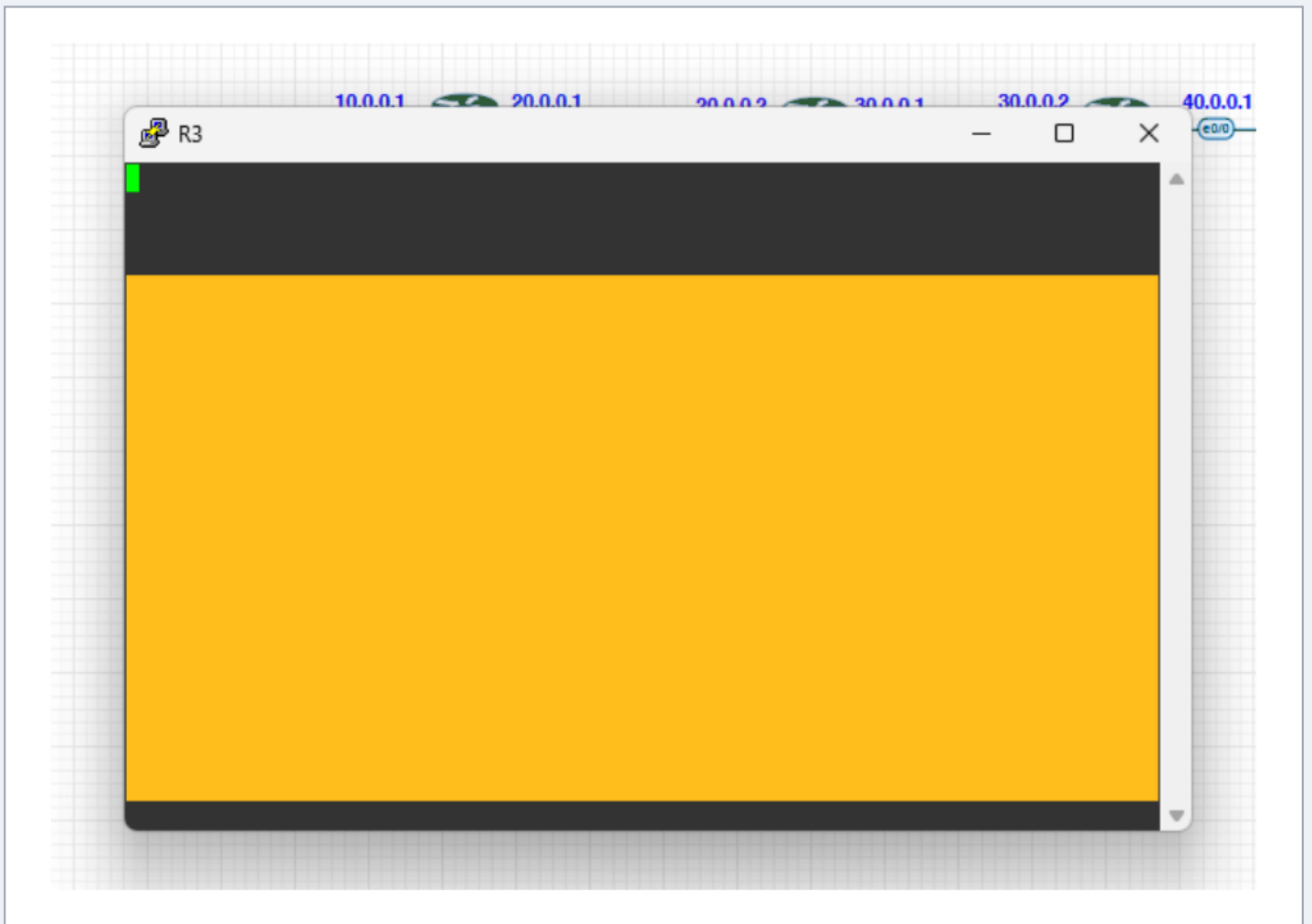
Router(config)#interface ethernet0/1
      ^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
*Aug 15 15:01:50.824: %SYS-5-CONFIG_I: Configured from console by console
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface ethernet0/1
Router(config-if)#ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
*Aug 15 15:03:00.785: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Aug 15 15:03:01.785: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
```

R2's Ethernet0/0 is set to 20.0.0.2 and Ethernet0/1 to 30.0.0.1.

```
Router(config-if)#exit
Router(config)#
Router#
*Aug 15 15:03:17.144: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              20.0.0.2        YES manual up          up
Ethernet0/1              30.0.0.1        YES manual up          up
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Router#
```

Both R2 interfaces confirmed up: 20.0.0.2 and 30.0.0.1.



R3's terminal is initializing before configuration begins.

```
Ethernet0/2          unassigned      YES unset  administratively down down
Ethernet0/3          unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip address 30.0.0.2 255.0.0.0
      ^
% Invalid input detected at '^' marker.

Router(config)#interface ethernet0/0
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Aug 15 15:07:55.182: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Aug 15 15:07:56.182: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/0, changed state to up
Router(config)#ip address 40.0.0.1 255.0.0.0
      ^
% Invalid input detected at '^' marker.

Router(config)#interface ethernet0/0
Router(config-if)#ip address 40.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#sho
*Aug 15 15:09:28.263: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              40.0.0.1        YES manual up          up
Ethernet0/1              unassigned      YES unset  administratively down down
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface ethernet0/1
Router(config-if)#ip address 30.0.0.2 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
*Aug 15 15:10:37.570: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Aug 15 15:10:38.570: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/1, changed state to up
Router(config)#exit
Router#
*Aug 15 15:10:43.263: %SYS-5-CONFIG_I: Configured from console by console
Router#
```

R3's Ethernet0/0 is set to 40.0.0.1 and Ethernet0/1 to 30.0.0.2.

```
Router#  
*Aug 15 15:10:43.263: %SYS-5-CONFIG_I: Configured from console by console  
Router#show ip interface brief  
Interface                IP-Address      OK? Method Status      Protocol  
Ethernet0/0               40.0.0.1        YES manual up          up  
Ethernet0/1               30.0.0.2        YES manual up          up  
Ethernet0/2               unassigned      YES unset  administratively down down  
Ethernet0/3               unassigned      YES unset  administratively down down
```

Both R3 interfaces confirmed up: 40.0.0.1 and 30.0.0.2.

```

C:\WINDOWS\system32\cmd.exe
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 40.0.0.2
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 40.0.0.1

C:\Documents and Settings\user>ping 40.0.0.1

Pinging 40.0.0.1 with 32 bytes of data:

Reply from 40.0.0.1: bytes=32 time=1ms TTL=255
Reply from 40.0.0.1: bytes=32 time<1ms TTL=255
Reply from 40.0.0.1: bytes=32 time=1ms TTL=255
Reply from 40.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 40.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>ping 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Reply from 30.0.0.2: bytes=32 time<1ms TTL=255
Reply from 30.0.0.2: bytes=32 time=1ms TTL=255
Reply from 30.0.0.2: bytes=32 time=1ms TTL=255
Reply from 30.0.0.2: bytes=32 time<1ms TTL=255

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\user>
  
```

Win-2 pings its gateway 40.0.0.1 and neighbor 30.0.0.2 successfully.

```
Router>
Router>
Router>
Router>enable
Router#show ip interface brief
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              10.0.0.1        YES manual up            up
Ethernet0/1              20.0.0.1        YES manual up            up
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2
Router(config)#exit
Router#
*Aug 15 15:19:36.444: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       30.0.0.0/8 [1/0] via 20.0.0.2
Router#
```

R1 gets a static route to 30.0.0.0/8 via 20.0.0.2, so it can reach R2's far network.

 R1

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    30.0.0.0/8 [1/0] via 20.0.0.2
```

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.2

Router(config)#exit

Router#s

*Aug 15 15:23:33.665: %SYS-5-CONFIG_I: Configured from console by console

Router#show ip route

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    40.0.0.0/8 [1/0] via 20.0.0.2
```

Router#

R1 also gets a route to 40.0.0.0/8 via 20.0.0.2 — R1's routing table is now complete.

```
R2
Router>
Router>enable
Router#show ip brief
      ^
% Invalid input detected at '^' marker.
Router#show interface brief
      ^
% Invalid input detected at '^' marker.
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
-----
Loopback0                 20.0.0.2        YES manual up          up
Ethernet0/0                30.0.0.1        YES manual up          up
Ethernet0/1                unassigned      YES unset  administratively down down
Ethernet0/2                unassigned      YES unset  administratively down down
Ethernet0/3                unassigned      YES unset  administratively down down

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
Router(config)#exit
Router#
*Aug 15 15:27:55.724: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
```

R2 gets routes to 10.0.0.0/8 (via 20.0.0.1) and 40.0.0.0/8 (via 30.0.0.2).

```
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
Router(config)#ip route 40.0.0.0 255.0.0.0 30.0.0.2
Router(config)#exit
Router#
*Aug 15 15:27:55.724: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/0
L    20.0.0.2/32 is directly connected, Ethernet0/0
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.1/32 is directly connected, Ethernet0/1
S    40.0.0.0/8 [1/0] via 30.0.0.2
Router#
```

R2's routing table now covers all four networks in the lab.

```
R3
Router>
Router>
Router>
Router>enable
Router#show ip brief
      ^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
Ethernet0/0    40.0.0.1        YES manual up          up
Ethernet0/1    30.0.0.2        YES manual up          up
Ethernet0/2    unassigned      YES unset  administratively down down
Ethernet0/3    unassigned      YES unset  administratively down down
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.0.0.0 30.0.0.1
Router(config)#ip route 20.0.0.0 255.0.0.0 30.0.0.1
Router(config)#exit
Router#sh
*Aug 15 15:30:47.523: %SYS-5-CONFIG_I: Configured from console by console
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 30.0.0.1
S    20.0.0.0/8 [1/0] via 30.0.0.1
C    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L    30.0.0.0/8 is directly connected, Ethernet0/1
L    30.0.0.2/32 is directly connected, Ethernet0/1
C    40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    40.0.0.0/8 is directly connected, Ethernet0/0
L    40.0.0.1/32 is directly connected, Ethernet0/0
Router#
```

R3 gets routes to 10.0.0.0/8 and 20.0.0.0/8 via 30.0.0.1 — its routing table is complete.

```

C:\Documents and Settings\User>ping 30.0.0.2
Pinging 30.0.0.2 with 32 bytes of data:
Reply from 30.0.0.2: bytes=32 time=1ms TTL=255
Reply from 30.0.0.2: bytes=32 time=1ms TTL=255
Reply from 30.0.0.2: bytes=32 time<1ms TTL=255
Reply from 30.0.0.2: bytes=32 time=1ms TTL=255

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\User>ping 10.0.0.1
Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=2ms TTL=253
Reply from 10.0.0.1: bytes=32 time=1ms TTL=253
Reply from 10.0.0.1: bytes=32 time=1ms TTL=253
Reply from 10.0.0.1: bytes=32 time=1ms TTL=253

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 3ms

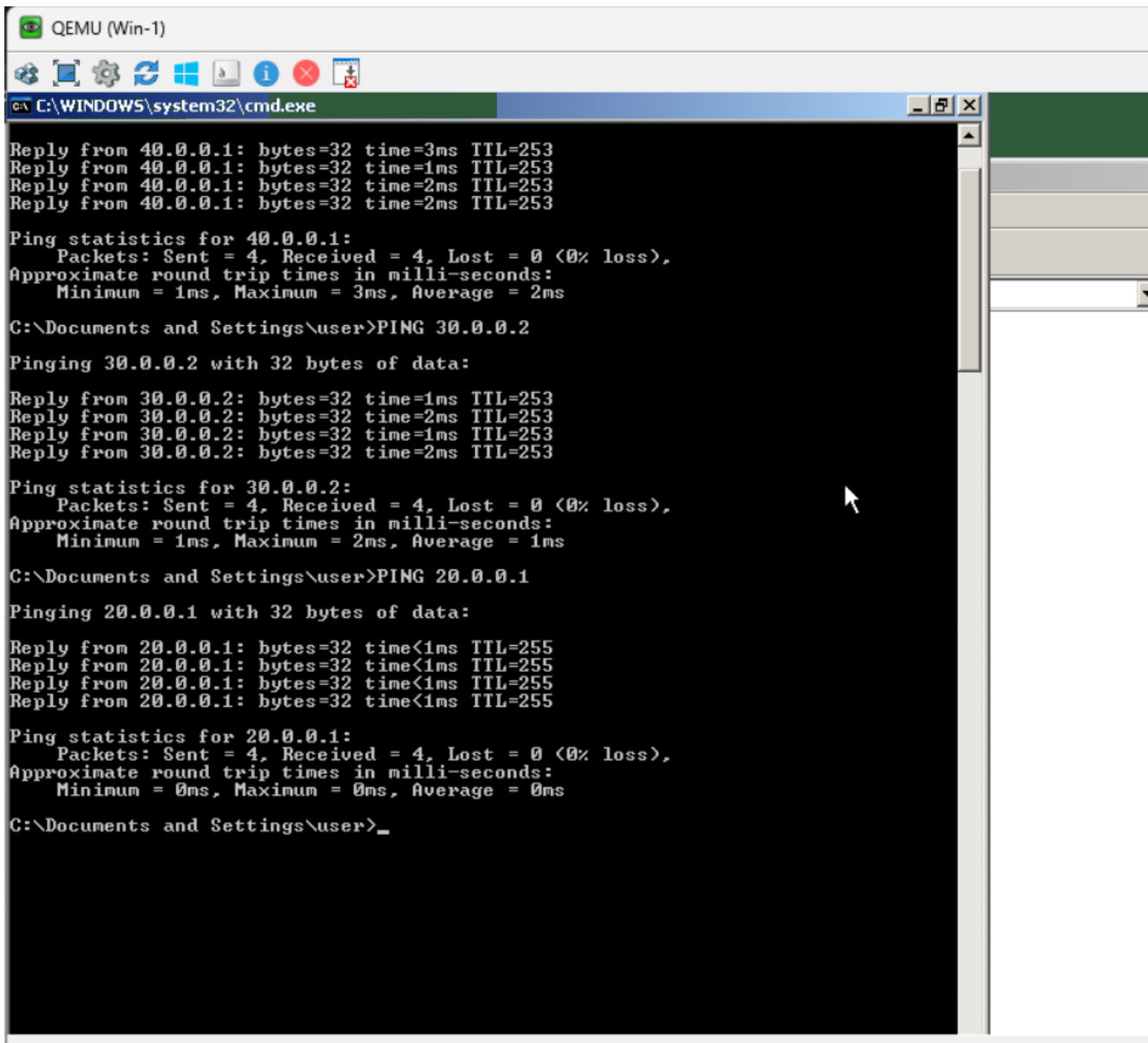
C:\Documents and Settings\User>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=2ms TTL=125
Reply from 10.0.0.10: bytes=32 time=1ms TTL=125

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>ping 20.0.0.1
Pinging 20.0.0.1 with 32 bytes of data:
Reply from 20.0.0.1: bytes=32 time=1ms TTL=253
Reply from 20.0.0.1: bytes=32 time=1ms TTL=253
Reply from 20.0.0.1: bytes=32 time=1ms TTL=253
Reply from 20.0.0.1: bytes=32 time=1ms TTL=253

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
  
```

Win-2 successfully pings 30.0.0.2, 10.0.0.1, Win-1 (10.0.0.10), and 20.0.0.1.



```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe

Reply from 40.0.0.1: bytes=32 time=3ms TTL=253
Reply from 40.0.0.1: bytes=32 time=1ms TTL=253
Reply from 40.0.0.1: bytes=32 time=2ms TTL=253
Reply from 40.0.0.1: bytes=32 time=2ms TTL=253

Ping statistics for 40.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>PING 30.0.0.2

Pinging 30.0.0.2 with 32 bytes of data:

Reply from 30.0.0.2: bytes=32 time=1ms TTL=253
Reply from 30.0.0.2: bytes=32 time=2ms TTL=253
Reply from 30.0.0.2: bytes=32 time=1ms TTL=253
Reply from 30.0.0.2: bytes=32 time=2ms TTL=253

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>PING 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\user>_
```

Win-1 successfully pings 40.0.0.1, 30.0.0.2, and 20.0.0.1 — both PCs can now reach each other end to end.